# Intensity, decibels and Doppler

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish intensity change from frequency shift.

**SUCCESS CRITERIA - I can...**

1. Use an intensity ratio separately from pitch.
2. Interpret a logarithmic sound-level difference.
3. Predict approach and recession Doppler trends.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “intensity” and “decibel” to explain your answer. Sentence starters are optional.

**intensity**: power passing through a unit area

**decibel**: a unit for a logarithmic sound-level comparison

**Doppler shift**: a frequency change associated with relative motion

## 01 Warm-up: bring back an earlier idea

A **500 Hz** tone lasts **0.3 s**. **Find number of cycles** and period.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

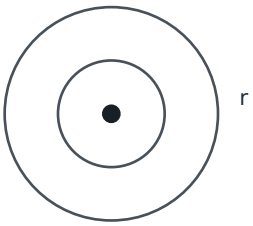
## 02 Worked example: explain the prediction

**Reference:** $\Delta\beta = \mathbf{10}\ \log_{10}(I_2/I_1)$.
I is intensity; β is sound level in dB. Tenfold intensity gives **+10 dB**. Approach generally raises observed frequency; recession lowers it.

**Worked reasoning:** For the same point source, doubling distance spreads its power over four times the area, so ideal intensity drops to one quarter. This assumes spherical spreading without extra absorption.

**Your turn:** Draw two concentric wavefronts and indicate why the outer surface shares the power more widely.

**Sentence starter:** For this answer, I treated ... as true. I found ... . This supports ... because ... .




Label distances and compare intensity.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

A moving source approaches a stationary listener, then recedes. **Sketch closer wavefront spacing** in front and wider spacing behind; **compare heard pitch**.

**Sentence starter:** Both ... . They differ in ... because ... .

Source moves right. Add wavefronts.



**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A sound meter records an intensity increase by a factor of **4**. Use $\Delta\beta = \mathbf{10}\ \log_{10}(I_2/I_1)$ to **find the level change**. Then **compare the intensities of 60 dB** and **80 dB** sounds. **Complete the ratio table**.

**Sentence starter:** Both ... . They differ in ... because ... .

| Case | Intensity ratio | Level change (dB) |
|---|---|---|
| A | ____ | ____ |
| B | ____ | ____ |

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

A loud stationary siren and a quiet approaching siren are compared. Does “louder” alone mean “higher frequency”? Explain **0**.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3

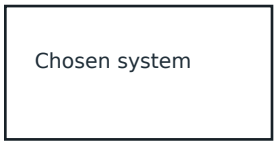
## 06 Review: connect an earlier idea

A **4 kg** sample with c=**500 J**/(kg·°C) warms **6°C**. **Find absorbed energy**. If it is **5%** of source energy, find source energy.
**Label energy entering**, leaving and stored.

**Word help:** absorbed = taken into the specified object or system

**Sentence starter:** I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.




Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 07 Exit check

Stationary listener; source approaches at **20 m/s**; emits **500 Hz**; $v_{sound}$=**340 m/s**.
1. Use $f_{obs}=f_s\ v/(v-v_s)$.
2. **Find observed frequency**.

**Sentence starter:** I found … [unit]. This value means … in this situation.

Source moves right. Add wavefronts.



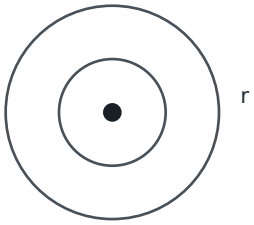
**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

Using $\Delta\beta=$**10** $\log_{10}(I_2/I_1)$, find the level change when distance triples from an ideal point source. Separately explain the higher observed pitch of an approaching source.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .




Label distances and compare intensity.

**Marzano check - circle:** 1 2 3